Exp 30 MATLAB CODE

```
clc;
clear;
close all;

% Cellular Frequency Planning and Handover Simulation

numCells = 7;
cellRadius = 1;
numUsers = 20;
numChannels = 21;

% Frequency Reuse Planning
frequencyGroups = [1 2 3 4 5 6 7];

figure;
hold on;

% Plot Hexagonal Cells
theta = 0:60:360;
xHex = cellRadius * cosd(theta);
yHex = cellRadius * sind(theta);

cellX = [0 1.5 1.5 0 -1.5 -1.5 0];
cellY = [0 0.866 -0.866 1.732 0.866 -0.866 -1.732];

for i = 1:numCells
    plot(cellX(i) + xHex, cellY(i) + yHex, 'b', 'LineWidth', 1.5);
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    text(cellX(i), cellY(i), ...
        sprintf('Cell %d\nF%d', i, frequencyGroups(i)), ...
        'HorizontalAlignment', 'center');
end

title('Cellular Frequency Planning');
xlabel('Distance X');
ylabel('Distance Y');
axis equal;
grid on;
hold off;

% User Mobility Simulation
time = 0:0.1:20;
userPosition = linspace(-4, 4, length(time));

baseStations = [-3 0 3];
signalStrength = zeros(length(time), length(baseStations));

for i = 1:length(baseStations)
    distance = abs(userPosition - baseStations(i));
    signalStrength(:,i) = 1 ./ (distance'.^2 + 0.5);
end

% Select Serving Cell
[~, servingCell] = max(signalStrength, [], 2);

% Handover Detection
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handoverCount = 0;
handoverTime = [];

for i = 2:length(servingCell)
    if servingCell(i) ~= servingCell(i-1)
        handoverCount = handoverCount + 1;
        handoverTime = [handoverTime time(i)];
    end
end

% Plot User Mobility and Signal Strength
figure;

subplot(3,1,1);
plot(time, userPosition, 'b', 'LineWidth', 1.5);
title('User Mobility');
xlabel('Time (seconds)');
ylabel('User Position');
grid on;

subplot(3,1,2);
plot(time, signalStrength, 'LineWidth', 1.5);
title('Received Signal Strength');
xlabel('Time (seconds)');
ylabel('Signal Strength');
legend('Base Station 1', 'Base Station 2', 'Base Station 3');
grid on;
```

```
subplot(3,1,3);
stairs(time, servingCell, 'r', 'LineWidth', 1.5);
title('Serving Cell During Mobility');
xlabel('Time (seconds)');
ylabel('Serving Cell');
yticks([1 2 3]);
grid on;

% Display Results
fprintf('Number of Cells = %d\n', numCells);
fprintf('Number of Users = %d\n', numUsers);
fprintf('Number of Frequency Channels = %d\n', numChannels);
fprintf('Total Handovers = %d\n', handoverCount);

disp('Handover Times (seconds):');
disp(handoverTime);
```

